Weather Shocks, Agriculture, and Crime: Evidence from India

David S Blakeslee* Ram Fishman†

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Abstract

We use detailed crime, agriculture, and weather data from India during the years 1971-2000 to conduct a systematic analysis of the relationship between weather shocks and multiple categories of crime. We find that drought and heat exert a strong impact on virtually all types of crimes, that the impact on property crimes is larger than for violent crimes, and that this relationship has been relatively stable over three decades of economic development. We then use seasonal and geographical disaggregations of weather and agricultural cultivation to examine the prevailing hypothesis that agricultural income shocks drive the weather-crime relationship in developing countries. The patterns we find are consistent with this hypothesis in the case of rainfall shocks, but suggest other mechanisms may play a more important role in driving the heat-crime relationship, consistent with evidence from industrialized countries.

Keywords: Crime, Income Shocks, Weather Shocks, Climate, Agriculture

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*New York University - Abu Dhabi. Email: davidsblakeslee@nyu.edu
†George Washington University. Email: rfishman@gwu.edu

1 Introduction

Growing interest in the potential impacts of climate change has spurred a substantial literature on the connection between weather conditions and conflict. This literature has now produced a substantial body of evidence that weather shocks increase civil conflict (Hsiang, Burke, and Miguel, 2013; Hsiang and Burke, 2014; Burke et al., 2009), which has dovetailed neatly with a related literature on the economic determinants of civil conflict (Collier and Hoeffler, 1998, 2004; Miguel, Satyanath, and Sergenti, 2004). A smaller number of papers have also shown that weather shocks can increase crime rates. Most of the latter research is based in industrialized countries, and has focused on the impacts of elevated temperatures,[1] with the few papers that are based in developing countries being focused on the impacts of rainfall shocks, which tend to be treated as operating primarily through economic channels (Miguel, 2005; Mehlum *et al.*, 2005; Sekhri and Storeygard, 2010).

Using spatially and temporally detailed data on weather, crime, and agricultural production from India over the three decades spanning 1971-2000, we conduct a systematic analysis of the connections between heat, drought and a range of property and violent crimes that is unique in its spatial and temporal scale; in the range of crimes investigated; and in the integrated analysis of potentially correlated temperature and rainfall variability.

We find strong evidence that both elevated heat and drought lead to substantial increases in crime rates of virtually all categories. During years in which rainy season precipitation is more than one standard deviation below the local long-term mean ("negative rainfall shock"), or temperatures one standard deviation above ("positive temperature shock"), crime rates increase by about 4% (an effect that is similar in magnitude to accumulated evidence from other studies reviewed by Hsiang *et al.*, 2013). While average crime rates have declined over time, the relationship between crime and weather has remained remarkably stable, despite the considerable economic development and structural change occurring across these years. Negative rainfall leads to an approximately 4% increase in crime across all three decades of our sample. The effect of temperature shocks declined somewhat between the 1970s and 1980s, but has since remained stable at about 4%. This stability suggests that economic development may have limited ability to shield vulnerable sectors of society from the warming and increased rainfall variability that is expected to occur over the coming

[1] Studies relating heat and crime are mostly based in the U.S. and include Anderson *et al.*, 2000; Auliciems and DiBartolo, 1995; Card and Dahl, 2011; Cohn and Rotton, 1997; Jacob *et al.*, 2007; Kenrick and MacFarlane, 1986; Larrick *et al.*, 2011; Mares, 2013; Ranson, 2012; Rotton and Cohn, 2000.

decades as a result of climate change.

We then go on to make use of the spatial and temporal precision of our data in order to make headway in understanding the mechanisms that might be driving the weather-crime relationship, an issue which remains a major gap in the literature (Hsiang, Burke, and Miguel, 2013). As mentioned above, previous studies of weather-crime connections in industrialized countries have tended to be confined to temperature fluctuations, and to explain their correlation with crime through a psychological mechanism relating heat to aggression. In contrast, studies of weather-crime connections in developing countries have tended to be confined to rainfall fluctuations, and to invoke to an income mechanism to explain their results. The latter hypothesis is a natural one given the sensitivity of agricultural productivity, wages, and employment to rainfall fluctuations in many developing countries, including in India (Guiteras, 2009; Fishman, 2012; Jayachandran, 2006; Kaur, 2012), and the large share of agriculture in both rural income and employment. It is also well grounded in the economic theory of crime, pioneered by Becker (1968), which predicts that individuals experiencing negative income shocks will be more likely to resort to criminal activity.[2]

The well established relationship between rainfall and agricultural income has led some researchers (e.g., Miguel, Satyanath, and Sergenti, 2004) to use rainfall shocks as an instrumental variable for estimating the causal relationship between income and civil conflict. While such a strategy is rendered more plausible by the lack of any clear alternative explanation for how drought might increase conflict, it's validity is nonetheless assumed rather than demonstrated. The causal channels linking temperature shocks and crime are more complicated, with temperatures having been shown to affect a wide range of economic and social outcomes (Dell *et al.*, 2014), including not only agricltural output (Deschênes and Greenstone, 2007; Lobell, Schlenker, and Costa-Roberts, 2011; Schlenker and Lobell, 2010;

[2]Beginning with Becker (1968), an extensive literature has invoked individual utility optimization to explain the economic factors driving the decision to engage in criminal activities. Particular emphasis is given to the opportunity costs of engaging in crime, which consist of the foregone income and other penalties in the event of capture, with the general prediction that reductions in legal income due to economic shocks will reduce the opportunity cost of crime and thereby increase its incidence. A number of studies have subsequently sought to empirically test the posited relationship between economic incentives and crime (see Freeman, 1999, for a review); and have generally found a relationship between economic distress and the incidence of crime (Gould, Weinberg and Mustard, 2002; Machin and Meghir, 2004; Grogger 1998). Virtually all of these studies are based on industrialized countries, however, with the result that little is known about how income shocks aff ect crime in the developing world, precisely those places most susceptible to highly disruptive economic shocks, and in which individuals are least able to insure themselves against large drops in income. Moreover, since economic conditions can themselves be influenced by the prevalence of crime (Bourguignon, 1999), and because unobservable characteristics such as institutions and culture may influence both the dependent and independent variables, identifying the causal relationship between the two is a perennial challenge.

Guiteras, 2009; Fishman, 2011, among others), but also labor supply and productivity in non-agricultural sectors (Hsiang, 2010; Dell *et al.*, 2012; Sudarshan and Tewari, 2013; Zivin and Neidel, 2014; Deryugina and Hsiang, 2014), and mortality (Burgess *et al.*, 2013), with much of this evidence coming from India. Temperature shocks are therefore more likely to operate on crime through multiple economic as well as physiological channels (Ranson, 2012), and it is not clear to what extent one should expect agricultural income to be the principal mediating factor between high temperatures and conflict in developing countries.

To make progress in distinguishing agriculture from other potential channels, we compare spatial and temporal hetergoenity in the patterns relating weather (temperature and precipitation) and crime, on the one hand, and those relating weather and agricultural production, on the other, and interpret parallels between these patterns as supportive of the agricultural income mechanism. Three types of heterogeneity are examined. The first makes use of the seasonal structure of the Indian agricultural calendar, wherein agricultural production depends most vitally on weather during the monsoon season. This approach contrasts with that of previous studies which have tended to rely on annual weather, thereby facilitating a closer analysis of the relevant mechanisms at play, and removing the noise introduced into the weather variables through the inclusion of weather events of little economic importance.[3] The second makes use of geographical variation in climatic conditions, including average precipitation and temperature, and the existence of a second monsoon in parts of southern and eastern India. The third compares property and violent crime rates, and tests the simple theoretical prediction that property crimes should be more strongly responsive to income shocks than violent crimes.

An examination of heterogeneities in the impacts of precipitation tends to support the agricultural income mechanism. Drought during the principal rainy season, the southwest monsoon, which is also the main cultivation period, has strong negative effects on agricultural production and strong positive effects on crime rates. Moreover, we find the same patterns in those parts of India which experience a second rainy season, the northeast monsoon. This monsoon sweeps along the eastern seaboard up through northeast India just as the summer monsoon is subsiding across much of the subcontinent. Here again, we find that when the northeast monsoon is weak, agricultural production suffers and crime rates increase. Geographical variability in mean rainfall provides further evidence in support

[3]In results not shown, we find that the estimated impacts of weather shocks on crime are reduced by the use of the annual, as opposed to seasonal, measures of weather variation.

of the agricultural channel. The substantial spatial variation in monsoon rainfall leads to some areas' having wetter climates than others, and a corresponding smaller sensitivity to reductions in rainfall. Insofar as increases in crime are indeed driven by changes in agricultural incomes, it stands to reason that areas with higher mean rainfall should, all things equal, display a smaller increase in crime during years of negative rainfall shocks. Consistent with this hypothesis, we find that areas with above average mean rainfall experience both a smaller decline in agricultural output during negative rainfall shocks, and a smaller increase in crime. Finally, a comparison of the effects of rainfall shocks on property and violent crimes reveals a statistically significant difference, with property crimes responding more strongly to negative rainfall shocks.

The parallel evidence in the case of temperature shocks is considerably more mixed, and suggests that agricultural income is not the sole, or even dominant, mechanism. On the one hand, high temperature shocks that occur during the agriculturally important rainy season (*kharif*), which lead to substantial declines in agricultural output, also lead to increases in crime. In addition, temperature shocks that occur during equally hot periods of the secondary agricultural season (*rabi*) have a negative effect on negative effect on *rabi* agricultural output, and a positive, but smaller, impact on crime (though the difference is not precisely estimated), which is consistent with the smaller extent of cultivation occurring in that season. At odds with a predominantly agricultural interpretation, however, is the fact that temperature shocks occurring during relatively cooler parts of the year do not affect crime despite causing a reduction in agricultural production, a finding that may favor mechanisms which are amplified in their effect during hotter parts of the year (such as psychological responses). Furthermore, and in clear contrast to rainfall, when we compare high temperature impacts in the hotter vs colder halves of India, the differential responsiveness of agricultural output is not accompanied by a differential responsiveness of crime. On the other hand, temperature shocks also tend to impact property crimes more highly than violent crimes, though the difference is smaller than that for rainfall, and in this case marginally insignificant (p-value=0.12).

One additional, particularly suggestive piece of evidence for the presence of additional channels of influence other than agricultural income is in the *relative* effects of temperature shocks, where positive temperature shocks are associated with an increase in crime similar in magnitude to that found with negative rainfall shocks, despite that latters' having a negative effect on agricultural output (-20.9%) roughly three times the size of the former (-6.7%).

This discrepancy is potentially explicable in terms of additional, non-agricultural impacts of high temperatures, including reductions in labor productivity in manufacturing recently found in India (Sudarshan and Tewari, 2013), and the heightened aggression described in the psychological literature and rigorously established in recent large scale economteric analyses (see Anderson, 2000, and Hsiang *et al.*, 2013, for a summary of this literature).

In summary, our examination of heterogeneities in the impacts of weather shocks on crime and agricultural output provides evidence which is consistent with a predominantly agricultural channel in the case of rainfall shocks, as posited by earlier research, but which also suggests that temperature shocks may be affecting crime rates through a variety of mechanisms, with agricultural income not necessarily foremost among them. Though none of the above pieces of evidence definitively resolves the operative mechanisms for rainfall and temperature shocks, nor rules out additional, related channels such as state capacity and political conflict, the cumulative evidence we present constitutes an important step in our understanding of this topic by both validating the importance of agricultural incomes, and revealing its limits in explaining the relationship between weather shocks and conflict.

The remainder of the paper is organized as follow. Section 2 describes the data used in this paper. Section 3 presents results relating weather and crime. Section 4 investigates the role of income shocks in mediating the weather-crime relationship. Section 5 discusses the results. Section 6 discusses climate change impacts, and section 7 concludes.

2 Data

Data on crime rates was obtained from India's National Crime Records Bureau (INCRB), housed under the Ministry of Home Affairs. INCRB produces annual documents on national and sub-national crime trends, including detailed statistics on the incidence of various crimes at the district levels, beginning in 1971.[4] The data provides a rich accounting of a wide variety of crimes. Among the crimes included are burglary, robbery, banditry, theft, riots, murder, rape, kidnapping, cheating, counterfeiting, and homicide. We measure crime incidence as the logarithm of the number of incidents occuring per 100,000 people in each district and year (population data are extrapolated linearly from Indian census data). Figure 1 presents the geographical distribution of the average rates of the different crime

[4]Crime data is also available before 1971, but only at the state level.

categories in our data. Figure 2 presents plots of India-wide average crime rates over time, and table 1 presents some summary statistics. The three columns tabulate the incidence of the indicated crime across the three decades spanning 1971-2000, and show a general decline in the incidence of most property crimes, with substantial declines in burglary, banditry, thefts, and robbery.[5] Kidnapping was relatively stable across this period, and riots declined slightly. Murder and rape increased somewhat during this period.

Monthly weather data is based on gridded precipitation and temperature data produced by the Indian Meteorological Department (Rajeevan *et al.,* 2005; Srivastava *et al.,* 2009) and converted to district-wise figures by area-weighted averaging over grid points falling within a given district (see Fishman, 2012 for more details). Our interest in the linkage between weather, agriculture, and crime leads us to measure heat exposure through the use of Degree Days, a measure of heat considered to be more relevant for crop growth than average temperatures, and one which has been previously used in empirical studies of weather-agricutlure relationships, including in India (Schlenker *et al.*, 2006, Guiteras 2009, Fishman 20120). Degree Days are defined by:

$$DDS = \sum_{d} D(T_{avg,d})$$

$$D(T) = \begin{cases} 0 & \text{if} \quad T \leq 8^{o}C \\ T - 8 & \text{if} \quad 8^{o}C < T \leq 32^{o}C \\ 24 & \text{if} \quad T > 32^{o}C \end{cases}$$

Monthly weather is used to calculate seasonal weather by summing over the months occuring within each season. Almost all rainfall occurs during the south-western monsoon season (figure 3), which lasts from June to September, though some parts of southern and eastern India receive a substantial portion of their rainfall during a second monsoon season, the north-eastern monsoon, which occurs between October and December. The warmest parts of the year occur from April to September. We follow the terminology of the Indian meteorological department and denote June-September as the "monsoon" season, October-December as the "post-monsoon" season, and March-May as the "pre-monsoon" season.[6]

[5] Robbery is distinguished from theft in that it includes violence in the commission of the crime. Banditry is distinguished from robbery by its involving 5 or more individuals in the commission of the crime.

[6] This classification scheme is described at https://en.wikipedia.org/wiki/Climate_of_India, which gives useful details of the Indian seasonal calendar.

The main agricultural growing season in India, the *kharif*, occurs between June and December (the exact period depends on the specific crop), when rain-fed cultivation is possible; for the poor especially, who lack access to irrigation, this season is the primary source of agricultural income. Rainy season crops, of which rice is the predominant one, are typically planted in June-July and harvested as late as December. As a result, production is sensitive to monsoon precipitation and to both monsoon and post-monsoon heat. The secondary agricultural season, the *rabi*, occurs from December-January to as late as May, and there is also some summer (pre-monsoon) cultivation in much more limited parts of the country. These dry season crops, of which wheat is the predominant one, rely on monsoon precipitation for irrigation or soil moisture, but are especially sensitive to spring and summer (pre-monsoon) heat.

District level agricultural data on the area planted, production, and yields of various crops during the two agricultural seasons were obtained from the Indian Harvest database produced by the Center for the Monitoring of the Indian Economy (see Fishman, 2012, for a complete description). As shown in table 1, agricultural production improved considerably during this interval due to the widespread adoption of HYVs with the green revolution. Yields per hectare for both rice and wheat nearly double, and gross product shows an increase of a similar magnitude.

An important issue in empirical studies on India for which the unit of observation is the district is the substantial partitioning (and sometimes re- combination) of districts that has occurred over time. This creates myriad challenges for determining the correct assignment of data to observations, as well as for the correct identification of the relevant unit for capturing time-invariant unobservables. The most common approach is to fix the districts on a certain date, and then aggregate later partitions of the district into the original. In our analysis, we approach this issue as conservatively as possible, maintaining as independent districts the districts resulting from partitions. This has the advantage, at times, of increasing the sample size, but the disadvantage of effectively removing some districts from our sample when partitioning yields districts with few observations.[7] We consider this the appropriate approach, as the partitioning of districts is likely to yield significant changes in institutions and local governance, which would only partially be captured with the use of constant district fixed effects.

[7]For example, it is not uncommon for sub-districts to have only a few years of data, so that the fixed effects will absorb much of the relevant variation.

3 The Impact of Weather Shocks on Crime Rates

3.1 Empirical Strategy

There are two principal specifications used in this analysis: in the first, we pool together all crimes in a single regression; in the second we estimate the same specification separately for each crime, with fixed effects and time effects adjusted accordingly. The specification employed is the linear model with the outcome given as log crime:

$$ln(y_{cist}) = \alpha + \beta \cdot P_{it}^{+} + \gamma \cdot P_{it}^{-} + \mu \cdot T_{it}^{+} + \nu \cdot T_{it}^{-} + \kappa_t + \delta_{c,i} + \kappa_s \times f(year_t) + \varepsilon_i$$

The natural log of the incidence of crime c (per 100k people) in district i, state s, and year t is regressed on dummy variables for positive and negative precipitation (P^{+}, P^{-}) and temperature (T^{+}, T^{-}) shocks in year t, district i. The positive-shock dummy takes a value of 1 for rainfall (temperature) movements 1 standard deviation or greater above the long-run district mean and zero otherwise, and the negative-shock dummy taking a value of 1 for rainfall (temperature) movements 1 standard deviation or more below the long-run district mean. As explained above, we decompose annual weather into the three seasons of the monsoon, the pre-monsoon, and the post-monsoon, and include separate shock indicators for each of them.[8] District-crime fixed effects and year fixed effects are also included ($\delta_{c,i}$ and κ_t, respectively), as well as linear state time trends in order to separate out time-invariant district characteristics and sample-wide annual shocks, and linear state time trends to capture localized scular trends in productivity, crime, and weather that could potentially generate spurious corrleations between our variable of interest.[9]

We cluster error terms at the district level, following the approach adopted by Sekhri and Storeygard (2010) and Burgess *et al.* (2013), who argue that measurement errors are likely to be correlated within districts across time. We find that our results are unchanged when we cluster instead at the state-year level in order to account for spatial correlation in weather.

A small number of observations in our sample report zero crime rates. These observations occur at a frequency of less than 2% for most crimes,[10] but result in undefined values

[8] There is some debate on whether the appropriate measure of weather shocks is year-on-year changes or levels (Miguel and Satyanath, 2011; Ciccone, 2011). The approach adopted here is more akin to the latter, with shocks defined by deviations of rainfall and temperature from their long-run means.

[9] Specifications are also estimated included interactions of the year fixed effects and state time trends with the crime fixed effects, as well as quadratic time trends, which yield nearly identical results.

[10] The one exception to this is banditry, for which zero values are encountered 7% of the time.

for our independent variable, the logarithm of crime rates. In dealing with similar issues, some authors have adopted a Poisson specification, which is well adapted to count-variable outcomes and handles the zero-value problem, but suffers from other limitations when a large number of fixed effects are included in the regression, as is the case in this paper.[11] An alternative approach, adopted by Pakes and Griliches (1980), is to replace the outcome variable with zero where the crime incidence is zero, and then to include a dummy indicating the transformation. We note that the incidence of zero values in our sample (around 2%) is substantially lower than in Pakes and Griliches's (1980) sample (around 8%). This specification has the advantages of more easily handling the inclusion of district fixed effects, and yielding coefficients of more transparent interpretation, and we adopt it here.

As discussed above, simple theoretical considerations suggest income shocks should have a larger impact on property crimes. We will therefore separately estimate regressions for property and non-property crime rates. Of the crimes included in the data, we classify burglary, banditry, theft, robbery, and riots as property crimes, and murder, rape, and kidnapping as non-property crimes. Of these, kidnapping and rioting would seem to occupy an ambiguous place; however, closer scrutiny justifies our classification. Kidnapping, for example, is disproportionately targeted against women, for reasons not entirely, or even principally, economic.[12] Riots[13] are known to occur during times of economic duress, particularly in response to heightened food prices, and are often characterized by widespread looting.

3.2 Results

We begin by reporting results from the pooled crime regressions. Coefficients for rainfall and temperature shocks are reported in Table 2, with columns (1)-(2) reporting results that include all types of crime; columns (3)-(4) including only property crimes; and columns (5)-(6) only non-property crimes. For each sample, we estimate a specification that includes only weather shocks in the main agricultural season (monsoon), and another that includes weather shocks in the three seasons of the monsoon, the post-monsoon, and the pre-monsoon.

[11] Hausman *et al.* (1984) is an example of this approach. Sekhri and Storeygard (2012) also use this approach in their study of the effect of rainfall shocks on dowry deaths in India. The latter paper, it should be noted, dealt with a crime for which zero-outcomes occurred, according to our calculations, in approximately 16% of the sample.

[12] The data indicates that the 74 % of kidnappings are targeted against women, though this disaggregation is only reported after 1987. Though others have classified this as an economic crime, such a classification overlooks this important non-economic component.

[13] Note that these are not the Hindu-Muslim riots analyzed by Bohlken and Sergenti (2010).

Using the full sample, we find that negative rainfall shocks during the monsoon are associated with a 3.6% increase in the incidence of crime, positive temperature shocks with a 3.9% increase, and negative temperature shocks with a 2.6% increase. Temperature shocks occurring outside the monsoon season have a smaller effect on crime, with pre-monsoon positive temperature shocks leading to a 1.7% increase in crime.[14] In columns (3)-(6), we see that the point estimates are larger for property crimes than for non-property crimes: whereas negative rainfall shocks lead to a 4.5% increase in property crimes, they lead to a smaller 2.3% increase in non-property crimes, with the difference between the two being statistically significant at the 10% level. Similarly, whereas positive temperature shocks lead to a 4.5% increase in property crimes, they lead to a smaller 2.7% increase in non-property crimes, though the difference is measured less precisely (p-value=0.12). Finally, the effect of negative temperature shocks occurs only for property crimes, which is consistent with negative temperature shocks' operating through primarily economic channels, and lacking the other causal pathways, particularly psychological effects, associated with positive temperature shocks.

We next estimate the effects of weather shocks on each crime individually. In addition to the inherent intrigue of the effects for individual crimes, this specification is important for two reasons. First, it allows us to assess whether the differential effects for property versus non-property crimes is sensitive to the classification adopted for kidnapping and riots, which, as noted in the introduction, are of somewhat ambiguous economic content. Second, as discussed in a number of studies, crime data can suffer from significant under-reporting.[15] To address this potential source of bias, researchers have focused on crimes less susceptible to under-reporting, particularly murder and robbery (Fajnzylber *et al.*, 1998). We will similarly spotlight the effects of weather shocks for these latter two types of crime to validate the overall findings.

Table 3 reports full individual crime regression estimates and figure 4 displays the estimated coefficients for monsoon period negative rain and positive temperature shocks. While four of the five property crimes shows a statistically significant increase with negative rainfall shocks, amongst non-property crimes it is only murder that shows a statistically significant increase. As was found in the pooled regressions, negative rainfall shocks generally yield

[14] The difference between the monsoon and pre-monsoon positive temperature effects has a p-value=0.16.

[15] This under-reporting will not be a problem for our identification strategy, so long as the bias is not correlated with the weather shock variables. It is not implausible that local authorities would in fact increase the level of under-reporting at precisely those times when crime is most increasing, as after an adverse weather shock. However, such behavior would, if anything, bias our results downwards.

a larger increase in property crimes than non-property crimes: riots increase by a statistically significant 5.7%, burglary by 5.3%, banditry by 4.8%, and robbery by 5.1%; whereas murder is the only non-property crime that shows a statistically significant increase (2.4%). Positive temperature shocks are associated with large increases in crime, particularly property crimes, with banditry increasing by 8.0%, riots by 5.7%, thefts by 5.2%, and robbery by 4.1%. Amongst the non-property crimes, murder shows a statistically insignificant increase of 2.1%, and rape a marginally significant 4.5% increase. Importantly, no differential effect is found for robbery and murder compared to other types of crime, indicating that under-reporting of crime is unlikely to be systematically biasing our results.

The relationship between weather shocks and crime has been estimated non-parametrically, due both to the demonstrated non-linear relationship between agricultural production and weather (see below), as well as non-linearities which would likely be present in other channels of influence (e.g., psychological responses to elevated temperatures). The weather dummies used above may collapse together distinct phenomena if, for example, abundant rainfall leads to bountiful harvests, but excessive rainfall to floods and destruction. We therefore estimate the relationship between weather shocks and crime using more disaggregated z-score intervals; Figures 5 and 6 show the results of these semi-parametric regressions. As can be seen, crime rises steadily with larger negative rainfall deviations, while positive deviations show little effect at any interval. The relationship between crime and temperature shocks is far noisier: while crime increases steadily with positive temperature shocks up to z-scores of 1.25-1.75, it drops precipitously thereafter, to become indistinguishable from zero. There is also an evident increase in crime with severe cold shocks.

4 The Role of Agriculture in Mediating the Weather-Crime Relationship

4.1 Agriculture and Weather Shocks

A central element of our analysis is an examination of the role agricultural productivity shocks may be playing in the observed weather-crime relationship. Weather shocks could also potentially operate through non-agricultural channels, such as disruptions to manufacturing, non-agricultural wages, psychological responses, or declines in state capacity. To examine the role of agriculture, we first turn to an analysis of the effect of weather shocks on agricultural production. Ideally, one would want to observe a distribution of annual income across a gradient of wealth. However, such data is unavailable at the required spatio-temporal resolution. Despite this limitation, previous studies have shown that agricultural output generally moves in tandem with agricultural wages and employment (Jayachandran, 2005; Kaur, 2012).

Various studies have already explored the relationship between weather and agricultural yields at some length (Guiteras, 2009; Fishman, 2012); here we provide a complementary analysis, by estimating the effects of the same vector of shocks whose impacts on crime were discussed above, and analyzing the correspondence of the two sets of impacts. If agricultural shocks play a significant role in the observed effect of weather on crime, one would expect those weather shocks that reduce agricultural output to increase crime, and those that do not affect agriculture to have a relatively smaller impact on crime (though the presence of other, non-agricultural mechanisms may still result in non-zero coefficients).

Table 4 reports the effects of the weather shocks on two measures of agricultural output: the total district output in 1000's of kgs ("product"), and the output in kgs per hectare of land ("yields").[16] For each of these outcome variables, we estimate the effects of the weather shocks on the "primary crop" grown in each district, where the primary crop is defined as that crop which occupies the largest portion of cultivated land across all years.[17] We choose to use the product of the primary crop over an attempt to estimate total revenue both because prices are endogenous, and because yield data is limited in the extent of crops covered, raising the concern that total revenue measures would be flawed. As discussed above, we estimate agricultural impacts separately for the primary rainy season (*kharif*) and the secondary, dry season (*rabi*). In column (1), we reproduce the results for crime rates to facilitate comparison.

Weather shocks are seen to have a large impact on crop production, particularly those shocks occurring during the monsoon period. Negative rainfall shocks lead to a 20.8% decline in agricultural product, and a 9% decline in dry season production. Positive temperature shocks during the monsoon lead to a 7.0% decline. These patterns are largely consistent with previous empirical analyses of weather-agriculture relationships in India (Guiteras, 2009; Fishman, 2012). As is apparent, the results are largely consistent with the crime

[16]We do not show the effect of climate on prices. This is because, with the advent of the railroad and the integration of agricultural markets, there is little within year price variability across districts (Donaldson, 2012). Confirming this finding, when we estimate the impact of climate on prices in our sample, we find no effect.

[17]The "primary crop" is determined as that which has the largest mean (log) area in a district over time.

results found earlier. Negative rainfall shocks and positive temperature shocks during the monsoon period decrease crop production and increase crime rates. The crime effect of negative temperature shocks is less readily explicable, as there is no corresponding reduction in agricultural output; however, we will see later that colder districts do in fact experience a decline in agricultural output during negative temperature shocks, and that the increase in crime is limited to precisely these districts.

We also see that the increase in crime with positive pre-monsoon temperature shocks corresponds to a 3% decline in dry season agricultural output, again giving evidence for an agricultural income mechanism being at play. The smaller extent of cultivation during the dry season can potentially help to explain why the effect of the pre-monsoon temperature shock on crime is smaller than that of the monsoon temperature shock. In results not shown, we find that the increase in crime occurs only in places where there is a larger extent of dry season crop cultivation, and is absent in districts with less dry season cultivation, though the difference between the two is measured imprecisely.

The apparent consistency of the effects of positive temperature shocks on agricltural production and on crime, for both the monsoon and pre-monsoon periods, belies a stark discrepancy in the magnitudes of these effects in comparison with those for rainfall. Specifically, whereas the effect of both positive temperature and negative rainfall shocks during the monsoon season leads to a very similar increase in crime, the effect of rainfall shocks on agricultural output is substantially larger (20.8% vs 7.0%). A natural explanation for this disparity is the potential presence of additional mediating links between heat and crime, including a psychological mechanism, in accord with a large body of research showing an association between high temperatures and elevated aggression (Anderson, 2001); or reductions in industrial output, including manufacturing (Hsiang, 2010; Jones and Olken, 2010; Dell *et al.*, 2012),[18] that are obeserved in India at both the plant and the worker level (Sudarshan and Tewari, 2013).

Another notable disagreement between the two patterns is the difference between the negative, albeit modest, impact of post-harvest temperature shocks on agriculture, and the lack of any apparent impact on crime rates.

[18]See Dell *et al.* (2014) for a review of this literature.

4.2 Agriculture and Weather Shocks: The Second Monsoon

To further test the posited agricultural mechanism for negative rainfall, we next turn to an analysis of the effect of India's second monsoon on the regions that it affects. Much of southern and eastern India experience a second monsoon season, which picks up in October, just as the summer monsoon is subsiding, and continues through November and some of December. This second monsoon season facilitates the cultivation of a second crop (without the necessity of using irrigation), which constitutes an important component of annual agricultural incomes in these areas.

In table 5, we estimate a specification including the second monsoon rainfall and temperature shocks, limiting the sample to those districts receiving more than 150 millimeters of rainfall during the second monsoon. Appendix figure A1 shows the spatial distribution of these districts, which are mostly concentrated along the eastern and southern coastal portions of the country.[19] The crop cultivated in these areas is almost exclusively rice, which is relatively water intensive, whereas the crop grown during this time in other areas tends to be wheat. As before, we estimate the effect of weather shocks on both crime and agriculture. While the summer monsoon weather shocks are no longer significant, the point estimates are similar to those found before, which is unsurprising given the considerably reduced sample size (73,361 vs 14,993). Importantly, we see that negative rainfall shocks during the second monsoon are associated with a 5.3% increase in crime, and decreases in both the rainy season monsoon crops (which are still being cultivated during this time) and the dry season crops.[20] The declines for the second monsoon crops are substantially larger than those for the first monsoon crop, which is consistent with the greater importance of rainfall during this season for the subsequent, second monsoon crop.

These results provide substantial evidence in favor of the importance of agricultural income in mediating the relationship between negative rainfall shocks and crime. As with monsoon shocks, the loss in agricultural income due to negative rainfall shocks is associated with a concomitant increase in crime. In this case, the increase in crime is, in fact, slightly larger than that for the monsoon season rainfall shock; and, despite the far smaller sample size, is estimated with a fair degree of precision.

[19]It should be noted that Kerala, one of the states experiencing the most rainfall during the second monsoon, is missing from our sample, due to the absence of the necessary population data.

[20]We retain the terminology of the dry season for consistency, though it is less appropriate in these regions.

4.3 Agro-Climatic Heterogeneities

To futher bolster the previous analysis on the role played by agricultural income in driving the relationship between weather shocks and crime, we next explore heterogeneities in treatment effects according to prevailing, district-level agro-climatic conditions. For this purpose, we disaggregate districts into those characterized mean rainfall (temperature) above and below the median across districts, and replicate the previous analysis, on the hypothesis that areas characterized by higher mean rainfall (temperature) conditions will be less (more) adversely affected by adverse shocks than those districts below the median.

Appendix figures A2 and A3 illustrate the logic of this approach. In figure A2 is shown the differential effect of rainfall deviations on agricultural output for low rainfall districts relative to higher rainfall districts; while in figure A3 is shown the differential effect for high temperature (degree days) districts relative to districts with lower mean temperatures.[21] As is apparent, districts with mean rainfall levels below the national median experience a steeper decline in agricultural output with negative rainfall shocks; while areas with high mean temperature levels (degree days) experience steeper declines with positive temperature shocks, and smaller declines with negative shocks. Columns (3)-(6) of panels A and B in table 6 give the corresponding regressions parameters, again demonstrating the heterogeneity in the effects of weather shocks across agro-climatic zones.

Columns (1)-(2) of panels A and B give the crime effects of rainfall and temperature shocks across the low and high mean rainfall and temperature samples, respectively. Looking first at the rainfall disaggregation, we can see that the crime results are broadly consistent with the agriculture effects. Specifically, the effect of negative rainfall shocks on agricultural output is roughly 1.5 larger in low-rainfall than high-rainfall districts, while the effect for crime is more than twice as high. While the difference between the crime coefficients is not statistically significant (p-value=0.19), the differential magnitudes of the coefficients is generally supportive of the posited agricultural channel.

In contrast, the results for temperature shocks across the two subsamples is only partly supportive of an agricultural channel. Specifically, the effect of positive temperature shocks on agricultural output is larger in high-mean temperature districts, while the effects on crime are virtually identical. This would seem to indicate that non-agricultural mechanisms

[21] Note the slight difference in the differentials used for the two graphs. This is because we expect *low* rainfall districts to be more sensitive to negative rainfall shocks than high rainfall districts, but *high* temperature districts to be more sensitive to positive temperature shocks than low temperature districts.

are at play, with psychological responses being the most obvious candidate. Interestingly, the effect of negative temperature shocks is more consistent with a primarily agricultural channel, with low-temperature districts experiencing both a decline in agricultural output, and an increase in crime, that are both statistically distinguishable from high-temperature districts. In results not shown, we also find that this differential crime effect exists only for property crimes, again consistent with negative temperatures' operating primarily through income channels. The correspondence between crime and agricultural impacts is weakest in the high temperature domain, which is precisely the domain where independent evidence exists for non-agricultural channels on income and crime.

4.4 Crime Elasticities

Using the negative monsoon rainfall results from table 4, the estimated elasticity of crime with respect to monsoon crop product is 0.169, and with respect to agricultural yield 0.258. In contast, had we instead used positive temperature shocks to estimate the elasticities, we would have identified crime elasticities of 0.557 and 1.258. The disparity in these estimated elasticities again highlights the inadmissability of mechanically treating weather shocks as interchangeable instrumental variables, and the necessity of coming to a deeper understanding of the mechanisms by which various weather shocks (and other income shocks) affect the incidence of conflict.

Because negative rainfall shocks are not likely to operate through any of the additional mechanisms cited above – having been found to have no effect on industrial production (Hsiang, 2010; Dell *et al.*, 2012) and unlikely to have important psychological effects – we would lean towards the elasticities estimated through this shock, and therefore to attribute the substantial majority of the temperature effect to non-agricultural channels, with the psychological effects seeming to us most plausible. Employing the elasticity estimated from the rainfall shock, we would in fact impute only 1.2 percentage points of the 3.9% increase in crime from positive temperature shocks to their effect on agriculture, and 2.7 percentage points to alternative channels, such as manufacturing and psychological responses.

5 Discussion

The results presented here, showing that the responses of crime to rainfall shocks closely mirror the relationship between the same shocks and agricultural revenue, constitute a significant contribution to the burgeoning literature on weather shocks and conflict. Using seasonally disaggregated weather and agricultural shocks, we find strong evidence that changes in agricultural production drive the relationship between negative rainfall shocks and crime. Simultaneously, a number of inconsistencies in the correspondence between temperatures, agricultural production, and crime indicate that high temperatures are likely affecting crime through additional, non-agricultural channels. These findings speak both to the burgeoning literature on weather-induced conflict, as well as to that on the economic determinants of crime, providing compelling evidence for theories emphasizing the cost-benefit calculus underlying the resort to illicit income generation, as well as the limitations of such an approach for understanding the effect of temperatures on conflict.

Though we have established a close connection between declines in agricultural output and increases in crime rates, data limitations prevent us from establishing the precise mechanism driving the relationship. Agricultural production shocks can affect the incomes of both land-owning farmers operating their own plots of land, as well as of landless casual wage laborers; and may also affect non-farming incomes through spillover to other sectors. Importantly, these income shocks are likely to most adversely affect the poorest farmers due to their lacking access to credit or other means of buffering themselves against adversity (Jayachandran, 2006); and it is precisely these individuals who, due to already low opportunity costs, are predicted to be most likely to adapt to negative income shocks through illicit income generation. Agricultural output shocks can also potentially increase crime through increases in food prices, which would reduce the real incomes of non-farming households; through spillovers to other sectors; or through reductions in state capacity, among others. We believe that the direct income channel is the most plausible, however, as India's agricultural markets have become increasingly integrated over the course of our study period, so that local agriculture shocks would not significantly affect prices, and agricultural output is only a marginal source of tax revenue for local governments, and therefore relatively unimportant for state capacity.

Despite the economic mechanisms underlying the relationship between weather shocks and crime, the fact that the relationship obtains even for crimes with less clear economic content suggests that there may be more a general social breakdown engendered by economic adversity not entirely explicable by cost-benefit analysis. This result is consistent across a number of countries and time periods, indicating that such non-economic mechanisms are widely prevalent. In our own results, we find that the expected differential effect for negative rainfall shocks across property versus violent crimes was positive and significant, but that the differential effect for positive temperatures was smaller and marginally insignificant.

6 Climate Change and the Evolution of Crime Responsiveness

We finally turn to an analysis of whether there have been changes over time in the effects of weather shocks on crime. The continuing build-up of atmospheric green house gases will increase India's temperatures in coming years, and significantly alter the timing of precipitation. Given our findings, there is the troubling possibility that these changes, in addition to their attendant economic disruptions, will also lead to increases in crime. Crime, in turn, entails both direct disruptions to human welfare, and possibly further reductions of income (Bourguignon, 1999). As we suggested earlier, however, there exist countervailing forces that may mitigate these effects; for, as countries undergo economic transformation, production becomes less dependent on agriculture, and agriculture itself becomes (potentially) less vulnerable to extreme weather, which should lower the responsiveness of crime to weather shocks.[22]

To explore this issue further, we examine how the responsiveness of crime has changed over the years of our data set. Because our data covers three decades, during which time there were dramatic changes in agricultural production, increasing urbanization, and significant economic growth, we can assess the effects of economic modernization on the observed relationship between weather and crime. This represents a significant contribution to the existing literature on climate and conflict, as most previous research has focused on economically static countries, so that it was not possible to identify how the estimated effects might change with economic and human development. Whether the remarkably consistent effects of climate on conflict found across a broad literature might be qualified by economic development is a question of considerable import (Hsiang *et al.*, 2013).

Figure 9 shows the relationship between crime and weather shocks across the three intervals: 1971-1980; 1981-1990; and 1991-2000. The effect of negative rainfall shocks is remarkably stable across time, remaining around 4% across these decades, with a slight dip in the 1980s. In contrast, the effect of temperature shocks drop from 8.8% in the 1970s,

[22] Additional salutary effects can come with economic development, as shown in Barreca *et al.* (2012), who find that mortality rates in the US have become less responsive to high temperatures due to the expansion of air conditioning.

down to 4.3% in the 1980s, then (a marginally insignificant) 3.1% in the 1990s. Table 7 gives the effects disaggregated by economic content. The decline of the temperature effect is found across both types of crime; however, with violent crimes showing a 5.3% response with temperature shocks in the 1970s, the declining effect had by the 1990s resulted in violent crimes' showing no statistically significant relationship with positive temperature shocks.

In appendix table A1, we disaggregate the decadal regressions across all types of crime. Burglary, robbery, riots, and kidnapping all continue to be highly responsive to negative rainfall shocks through the 1990s. The only crimes that clearly become less responsive to negative rainfall shocks are banditry and rape, though the latter is statistically insignificant in all decades, and subject to considerable reporting bias. Riots show a marked decline in their responsiveness to temperature shocks during the 1980s, though the effect re-emerges in part in the 1990s. Banditry, thefts, riots, and rape become less responsive to positive temperature shocks, while robbery and kidnapping become more responsive. Robbery and thefts seem to move in opposite directions, raising the possibility that there has either been a change in reporting conventions, with thefts being re-classified as robbery,[23] or a true change in the nature of property crime, with violence more likely to be deployed in the expropriation of property.

7 Conclusion

We provide some of the first evidence on the drivers of crime in developing countries, using a data set covering nearly a billion people, and spanning three decades of economic and social transformation. Variation in temperature and rainfall are found to play a large role in driving changes in the incidence of crime. These results obtain for both property and violent crimes, though the effects are larger, more robust, and remarkably uniform for the former. This fact, and the alignment of these effects with the responses of agricultural production to the same weather shocks provide strong evidence for the presence of an income mechanism, whereby falling agricultural output leads to increases in crime, due to the decline in opportunity costs associated with criminal activity.

However, we also find that the effect of elevated temperatures is higher than expected based on their effect on agricultural output, consistent with the presence of both non-agricultural income and psychological mechanisms, as found across a substantial literature

[23] Recall that the only difference between the two crimes is that robbery is a theft in which force is used.

on temperature. The effects of income shocks on crime are highly non-symmetric: while negative agriculture shocks consistently lead to increases in crime, positive agriculture shocks do no lead to a decline. In results not shown, we find that secular increases in rural income, however, due to the expansion of irrigation, are associated with significant reductions in crime, highlighting the importance of distinguishing between long-term and short-term income changes in empirical and theoretical research on the economic determinants of crime.

Despite higher incomes, greater access to consumption smoothing instruments, and reduced susceptibility of agriculture to climatic variability that accompany economic growth, there is little evidence that crime has become less responsive to extreme rainfall than it was prior to these improvements. This may be taken as evidence that, despite India's remarkable gains in human and economic development, the poorest members of society, being those most likely to resort to crime during times of economic adversity, continue to remain highly vulnerable to aggregate economic shocks. This observation adds to existing concerns about the equity of India's remarkable economic growth. However, the responsiveness of crime to elevated temperatures declined substantially between the 1970s and present day, though remained high for some of the most pernicious crimes, such as murder, kidnapping, and robbery. In sum, the evidence indicates that policy makers will have to be vigilant against rising crime rates as India's climate is increasingly buffeted by the global accumulation of atmospheric greenhouse gases.

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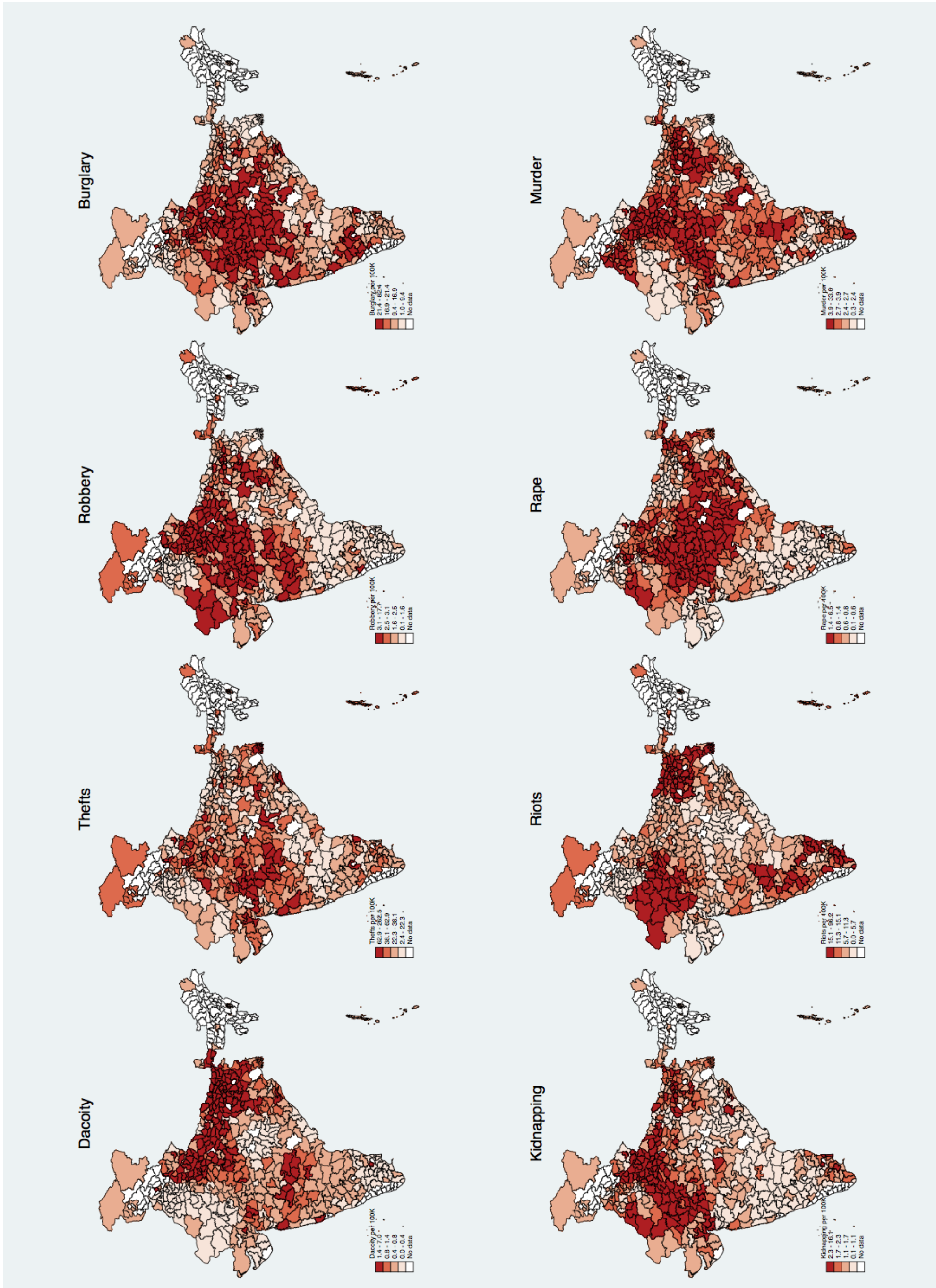

Figure 1: The spatial distribution of average 1971-2000 crime rates (crimes committed per 100,000 population).

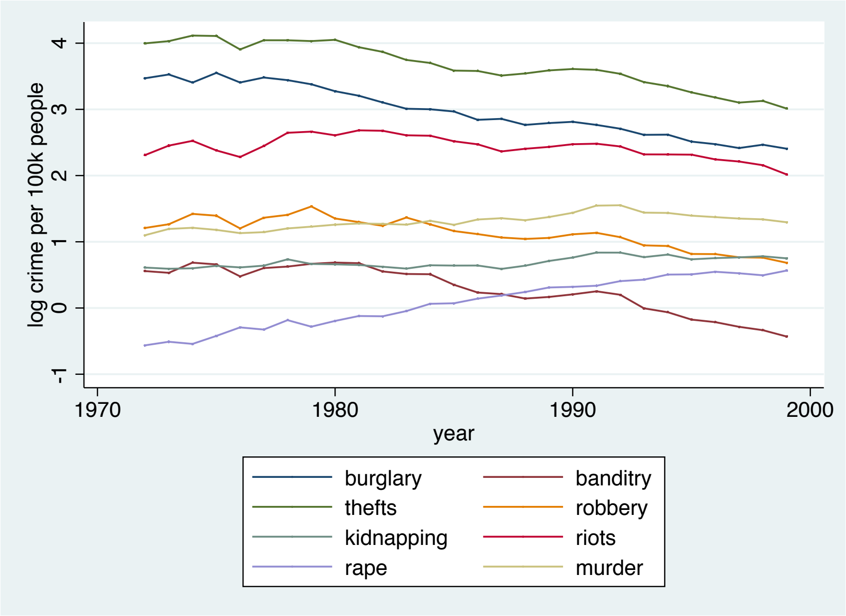

Figure 2: Plots of the (logarithm) of India-wide average crime incidence (per 100,000 population) over time.

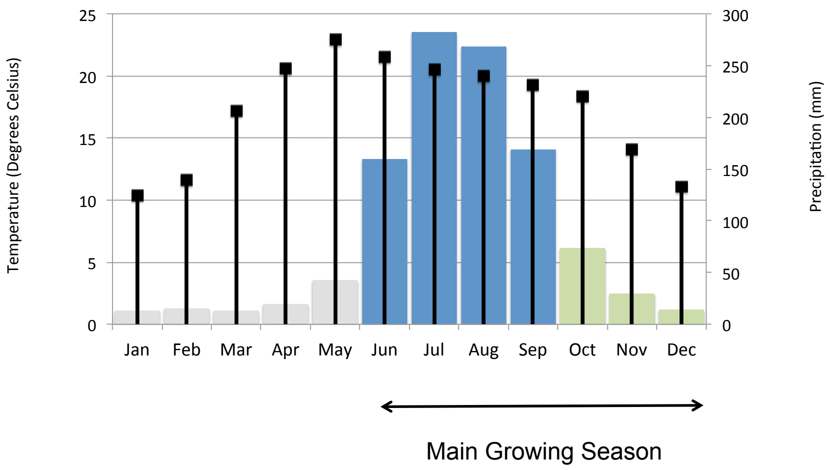

Figure 3: Average climate in India. Monthly average temperature (black drop lines) and precipitation (bars). Monsoon crops are usually sown June-September and harvested between October and December. The blue and green shades represent the breakup of weather indicators used in the empirical analysis.

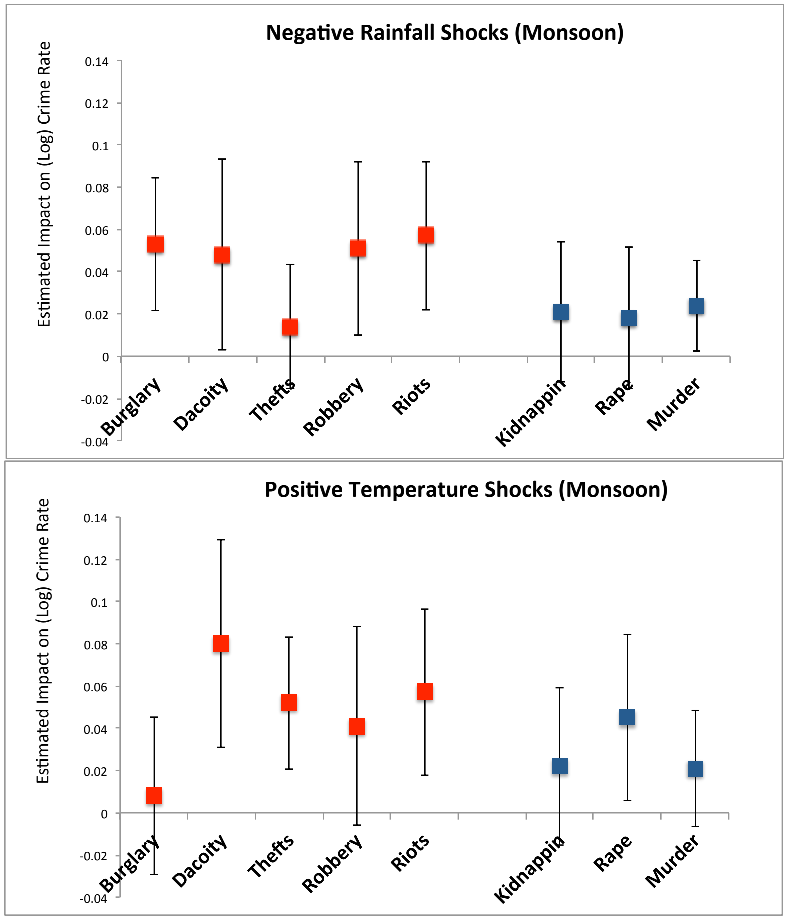

Figure 4: Estimated coefficients from regressions of disaggregated crime rates on negative monsoon rainfall shocks (top) and positive monsoon temperature shocks (bottom). Property crimes coefficients are indicated by red markers, and violent crime coefficients are indicated by grey markers.

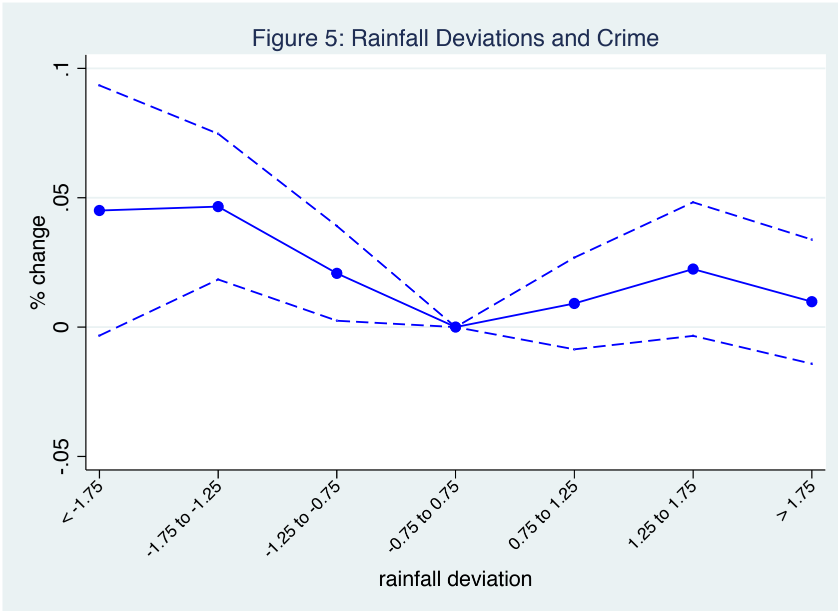

Figure 5: Coefficients from a regression of crime on monsoon rainfall in the indicated standard deviation bins.

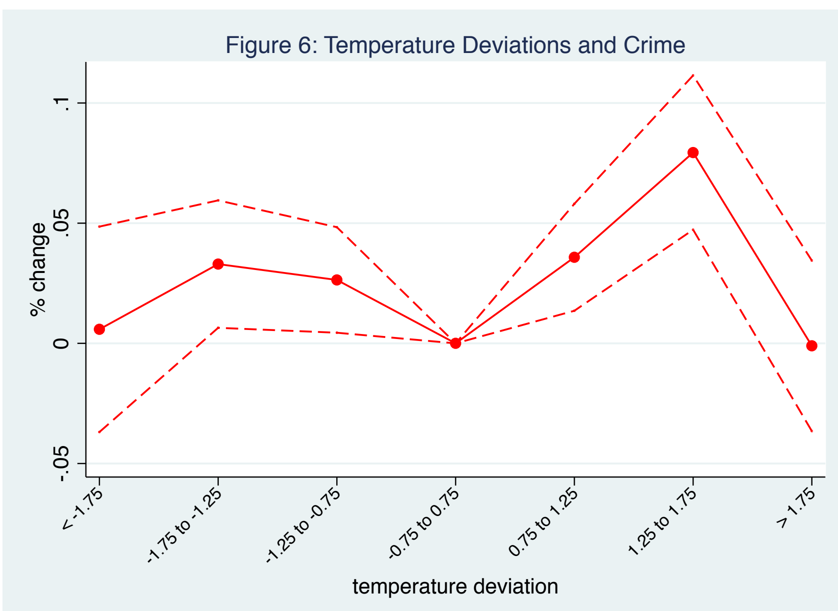

Figure 6: Coefficients from a regression of crime on monsoon temperature (dds) in the indicated standard deviation bins.

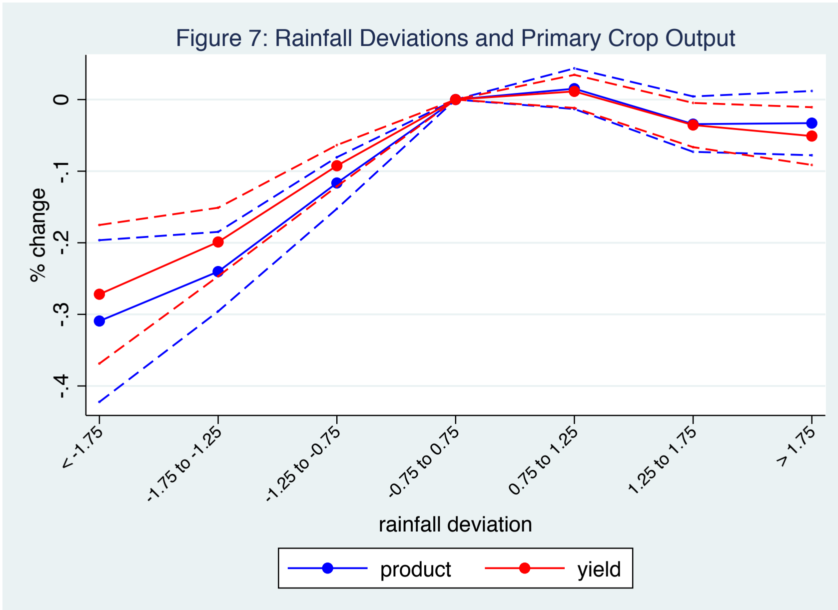

Figure 7: Coefficients from a regression of the monsoon agricultural revenue on monsoon rainfall in the indicated standard deviation bins.

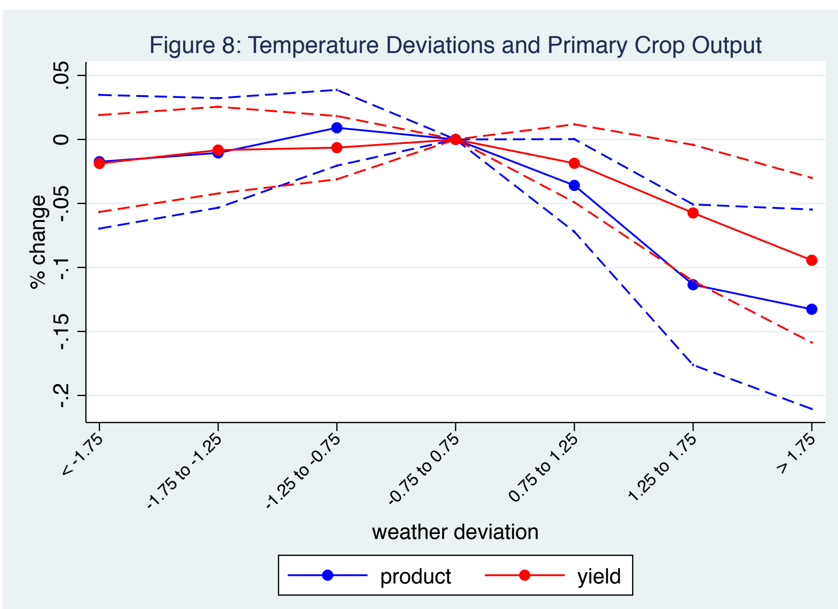

Figure 8: Coefficients from a regression of the monsoon agricultural revenue on monsoon temperature (dds) in the indicated standard deviation bins.

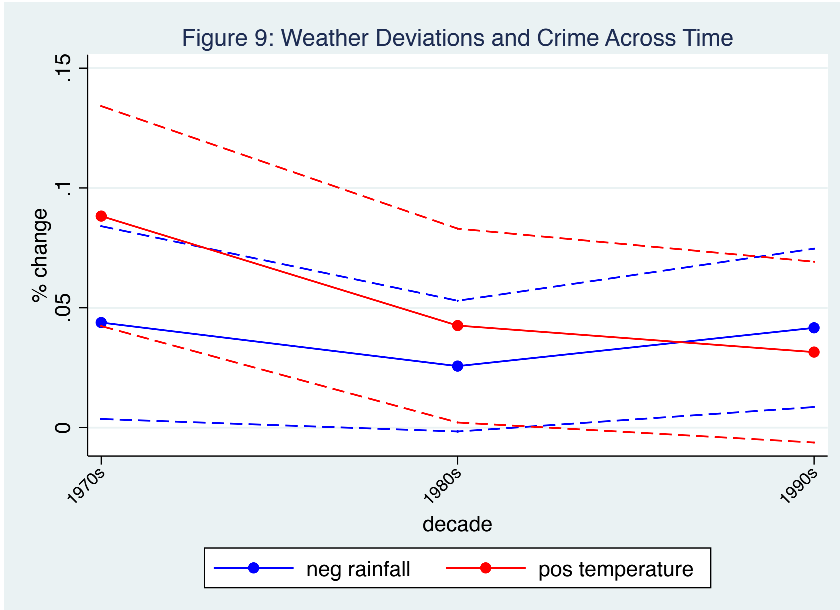

Figure 9: Coefficients from a regression of crime on monsoon rainfall and degree days across the indicated decades, with separate weather shocks dummies used for each decade.

Table 1: Summary Statistics

	all years (1)	1970s (2)	1980s (3)	1990s (4)
crimes per 100k				
property				
burglary	20.51	32.97	20.14	13.17
banditry	1.44	2.07	1.58	0.94
thefts	42.16	64.13	42.68	28.25
robbery	3.19	4.04	3.419	2.48
riots	11.59	11.85	12.77	10.46
non-property				
kidnapping	2.04	1.94	1.917	2.19
rape	1.20	0.68	1.07	1.613
murder	3.81	3.27	3.714	4.23
agricultural output				
irrigation	0.595	0.510	0.611	0.642
primary crop				
product	180.14	151.62	177.45	204.37
yield	1379.65	1083.88	1286.65	1691.58
area	142.42	140.42	146.90	139.78

Table 1 gives the summary statistics for crime and agriculture. Crime is given as incidents per 100k population. Irrigation is given as percentage of cultivate land irrigated. Product is thousands of kgs, yield is kgs per hectare, and area is in thousands of square kilometers.

Table 2: Weather Shocks and Pooled Crime

Outcome: log crime (100k)	full sample		property crimes		non-property	
	(1)	(2)	(3)	(4)	(5)	(6)
monsoon shocks						
neg rain	0.036***	0.036***	0.044***	0.045***	0.023**	0.023**
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
pos rain	0.008	0.008	0.009	0.008	0.007	0.006
	(0.009)	(0.009)	(0.010)	(0.010)	(0.010)	(0.010)
neg dds	0.026***	0.027***	0.039***	0.041***	0.005	0.006
	(0.010)	(0.010)	(0.012)	(0.012)	(0.010)	(0.011)
pos dds	0.040***	0.039***	0.045***	0.046***	0.031**	0.027**
	(0.012)	(0.012)	(0.014)	(0.014)	(0.013)	(0.013)
pre-monsoon shocks						
neg dds		0.007		0.006		0.009
		(0.010)		(0.012)		(0.011)
pos dds		0.017**		0.019**		0.012
		(0.008)		(0.009)		(0.010)
post-monsoon shocks						
pos dds		0.007		0.001		0.018
		(0.010)		(0.013)		(0.012)
neg dds		0.004		-0.005		0.018*
		(0.009)		(0.011)		(0.010)
R-squared	0.898	0.898	0.915	0.915	0.818	0.818
N	73361	73361	45789	45789	27572	27572

The outcome variable is crimes per 100k. In columns (1)-(2), all crimes are included; while in columns (3)-(4) only property crimes, and columns (5)-(6) non-property crimes. The climate shocks are defined as rainfall/temperature 1 standard deviation from the mean. Dummies are included for the district-crime. Year fixed effects are included, as well as state-year time tends. Error terms are clustered at the district level.

Table 3: Weather Shocks and Disaggregated Crime

Outcome: log crime (100k)	property crimes					non-property crimes		
	burglary	banditry	thefts	robbery	riots	kidnapping	rape	murder
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
monsoon shocks								
neg rain	0.053***	0.048**	0.014	0.051**	0.057***	0.021	0.018	0.024**
	(0.016)	(0.023)	(0.015)	(0.021)	(0.018)	(0.017)	(0.017)	(0.011)
pos rain	0.004	0.036*	0.005	0.003	-0.005	0.012	0.008	-0.004
	(0.012)	(0.019)	(0.011)	(0.016)	(0.017)	(0.014)	(0.016)	(0.010)
neg dds	0.027**	0.069***	0.027**	0.068***	0.013	0.009	0.005	0.005
	(0.013)	(0.025)	(0.011)	(0.020)	(0.019)	(0.016)	(0.020)	(0.011)
pos dds	0.008	0.080***	0.052***	0.041*	0.057***	0.022	0.045**	0.021
	(0.019)	(0.025)	(0.016)	(0.024)	(0.020)	(0.019)	(0.020)	(0.014)
pre-monsoon shocks								
neg dds	0.024*	0.036*	0.004	-0.018	-0.018	0.023	-0.020	0.019*
	(0.013)	(0.022)	(0.013)	(0.023)	(0.019)	(0.017)	(0.017)	(0.010)
pos dds	-0.019*	0.025	0.033***	0.052***	0.006	0.002	0.028*	0.005
	(0.010)	(0.023)	(0.010)	(0.018)	(0.015)	(0.015)	(0.017)	(0.010)
post-monsoon shocks								
neg dds	0.015	0.033	-0.000	-0.006	-0.042**	-0.002	0.022	0.022*
	(0.017)	(0.024)	(0.014)	(0.022)	(0.018)	(0.019)	(0.021)	(0.013)
pos dds	-0.000	-0.052**	-0.025*	0.024	0.025	0.014	0.022	0.009
	(0.014)	(0.024)	(0.013)	(0.019)	(0.017)	(0.016)	(0.017)	(0.011)
R-squared	0.837	0.790	0.826	0.775	0.806	0.743	0.781	0.762
N	9181	9126	9125	9181	9176	9194	9200	9178

The outcome variable is crimes per 100k. Regressions are estimated separately for each type of crime. The weather shocks are defined as rainfall/temperature 1 standard deviation from the mean. Dummies are included for the district-crime. Year fixed effects are included, as well as state-year time tends. Error terms are clustered at the district level.

Table 4: Weather Shocks, Agriculture, and Crime

	log crime	log agricultural output			
		monsoon crop		dry season crop	
		product	yield	product	yield
	(1)	(2)	(3)	(4)	(5)
monsoon shocks					
neg rain	0.036***	-0.208***	-0.169***	-0.094***	-0.040***
	(0.010)	(0.022)	(0.018)	(0.015)	(0.011)
pos rain	0.008	-0.013	-0.019	0.045***	0.023***
	(0.009)	(0.013)	(0.012)	(0.011)	(0.007)
neg dds	0.027***	-0.005	-0.014	0.020	0.002
	(0.010)	(0.016)	(0.012)	(0.012)	(0.008)
pos dds	0.039***	-0.070***	-0.031*	-0.022	0.011
	(0.012)	(0.022)	(0.018)	(0.015)	(0.010)
pre-monsoon shocks					
neg dds	0.007	0.075***	0.059***	0.019	0.012
	(0.010)	(0.015)	(0.013)	(0.013)	(0.009)
pos dds	0.017**	0.028*	0.012	-0.030**	-0.045***
	(0.008)	(0.015)	(0.012)	(0.014)	(0.010)
post-monsoon shocks					
neg dds	0.007	0.066***	0.077***	0.036**	0.004
	(0.010)	(0.017)	(0.013)	(0.014)	(0.010)
pos dds	0.004	-0.074***	-0.064***	-0.046***	-0.032***
	(0.009)	(0.019)	(0.016)	(0.016)	(0.010)
R-squared	0.898	0.912	0.842	0.957	0.893
N	73361	5958	5958	7398	7398

The outcome variable is (log) total product and yield of the primary crops grown in the monsoon and dry seasons, respectively, primary crop. The weather shocks are defined as rainfall/temperature 1 standard deviation from the mean. Dummies are included for the district. Year fixed effects are included, as well as state-year time tends. Error terms are clustered at the district level.

Table 5: Weather Shocks, Agriculture, and Crime: Second Monsoon

	log crime	log agricultural output			
		monsoon crop		dry season crop	
		product	yield	product	yield
	(1)	(2)	(3)	(4)	(5)
monsoon shocks					
neg rain	0.034	-0.248***	-0.153***	-0.197*	-0.042*
	(0.022)	(0.031)	(0.023)	(0.118)	(0.024)
pos rain	0.007	0.037	0.017	0.151*	0.007
	(0.016)	(0.024)	(0.021)	(0.071)	(0.026)
neg dds	0.017	0.028	0.007	-0.195*	-0.066*
	(0.029)	(0.047)	(0.039)	(0.103)	(0.034)
pos dds	0.029	-0.100**	-0.032	-0.193	0.022
	(0.039)	(0.038)	(0.031)	(0.129)	(0.038)
pre-monsoon shocks					
neg dds	0.010	0.017	-0.010	-0.039	0.012
	(0.016)	(0.039)	(0.032)	(0.085)	(0.027)
pos dds	0.005	-0.023	-0.001	0.021	-0.056*
	(0.017)	(0.027)	(0.022)	(0.115)	(0.029)
second monsoon shocks					
neg rain	0.053**	-0.095***	-0.064***	-0.225***	-0.084**
	(0.026)	(0.035)	(0.024)	(0.069)	(0.038)
pos rain	0.017	-0.024	-0.031	0.055	0.004
	(0.016)	(0.025)	(0.020)	(0.064)	(0.022)
neg dds	-0.001	0.054*	0.067***	0.102	0.026
	(0.030)	(0.029)	(0.022)	(0.121)	(0.031)
pos dds	0.035	-0.007	0.016	0.179	-0.018
	(0.025)	(0.034)	(0.023)	(0.124)	(0.023)
R-squared	0.908	0.925	0.825	0.819	0.861
N	14933	1319	1319	1206	1205

The outcome variable is (log) total product and yield of the primary crops grown in the monsoon and dry seasons, respectively, primary crop. The weather shocks are defined as rainfall/temperature 1 standard deviation from the mean. Dummies are included for the district. Year fixed effects are included, as well as state-year time tends. Error terms are clustered at the district level. The sample is limited to districts which receive the second monsoon.

Table 6: Weather Shocks, Agro-Climatic Disaggregations, and Crime

	log crime		log product		log yield	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Temperature Interactions						
neg dds	0.044***	0.005	-0.039**	0.020	-0.057***	0.015
	(0.013)	(0.016)	(0.020)	(0.024)	(0.015)	(0.018)
pos dds	0.057**	0.037**	-0.071**	-0.101***	-0.025	-0.057*
	(0.020)	(0.015)	(0.028)	(0.038)	(0.022)	(0.034)
R-squared	0.906	0.890	0.909	0.918	0.815	0.866
N	36565	36796	310	2871	3105	2871
Panel B: Precipitation Interactions						
neg rain	0.047***	0.021	-0.260***	-0.167***	-0.193***	-0.147***
	(0.016)	(0.013)	(0.030)	(0.029)	(0.026)	(0.025)
pos rain	0.024	-0.007	-0.010	-0.030	-0.017	-0.032*
	(0.015)	(0.010)	(0.020)	(0.019)	(0.016)	(0.017)
R-squared	0.903	0.893	0.905	0.923	0.876	0.759
N	36661	36700	3000	2976	3000	2976

The outcome variables are log total revenue and log crime. Panel A gives the results from from regressions disaggregating the sample according to the district-level temperature-sensitivity of agricultural revenue. Panel B gives the results from regressions disaggregating the sample according to mean monsoon rainfall levels. The weather shocks are defined as rainfall/temperature 1 standard deviation from the mean. Dummies are included for the district. Year fixed effects are included, as well as state-year time tends. Error terms are clustered at the district level.

Table 7: Weather Shocks and Crime, by Decade

Outcome: log crime (100k)	full sample		property crimes		non-property	
	neg rain	pos dds	neg rain	pos dds	neg rain	pos dds
	(1)	(2)	(3)	(4)	(5)	(6)
decade						
1970s	0.044**	0.088***	0.050**	0.109***	0.036*	0.052**
	(0.020)	(0.023)	(0.025)	(0.028)	(0.020)	(0.023)
1980s	0.026*	0.042**	0.035**	0.056**	0.010	0.021
	(0.014)	(0.021)	(0.017)	(0.025)	(0.016)	(0.024)
1990s	0.041**	0.031	0.049**	0.036*	0.029*	0.024
	(0.017)	(0.019)	(0.020)	(0.022)	(0.017)	(0.019)
R-squared	0.894		0.909		0.812	
N	74932		46772		28160	

The outcome variable is crimes per 100k. In columns (1)-(2), all crimes are included; while in columns (3)-(4) only propety crimes, and columns (5)-(6) non-property crimes. The climate shocks are defined as rainfall/temperature 1 standard deviation from the mean; dummies are included separately for shocks occurring in each decade. Dummies are included for the district-crime. Year fixed effects are included, as well as state-year time tends. Error terms are clustered at the district level.

Appendix

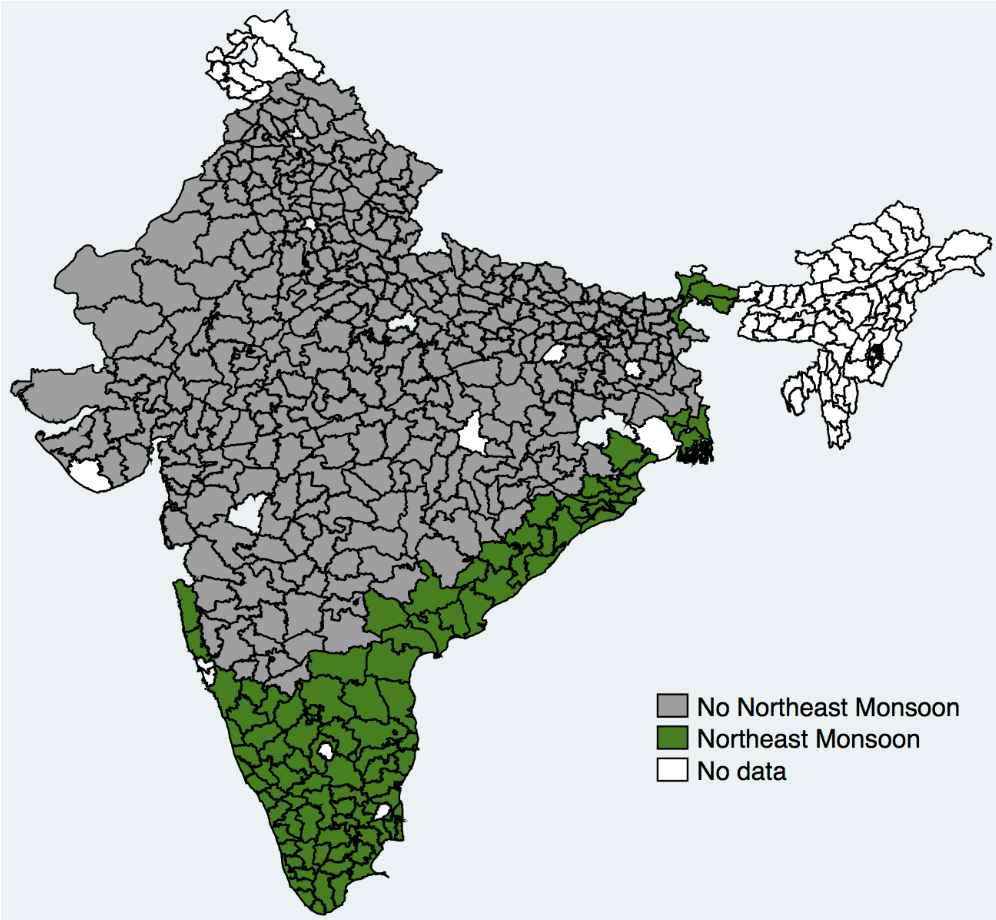

Figure A1: The spatial distribution of districts exposed to the second monsoon.

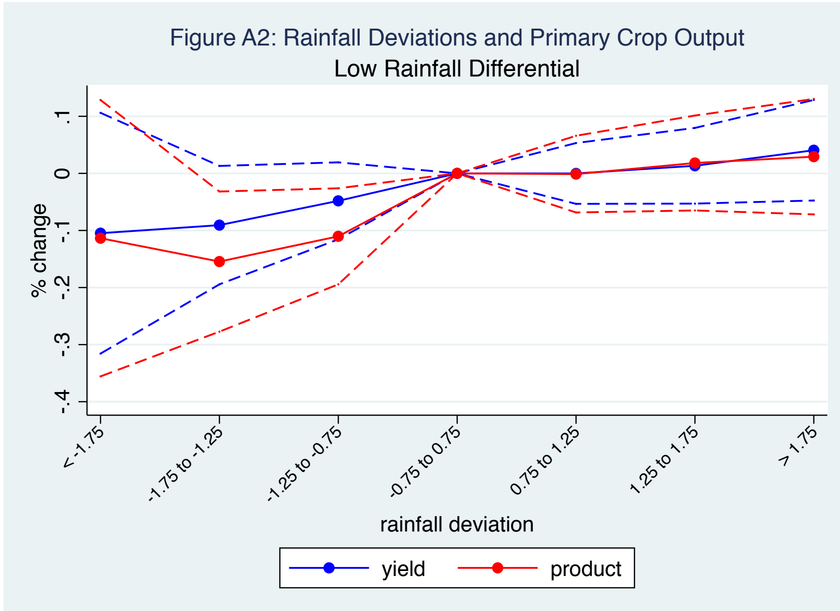

Figure A2: Interaction coefficients from a regression of the monsoon agricultural output on monsoon rainfall in the indicated standard deviation bins, where the interaction terms give the differential effect for shocks in low mean rainfall districts.

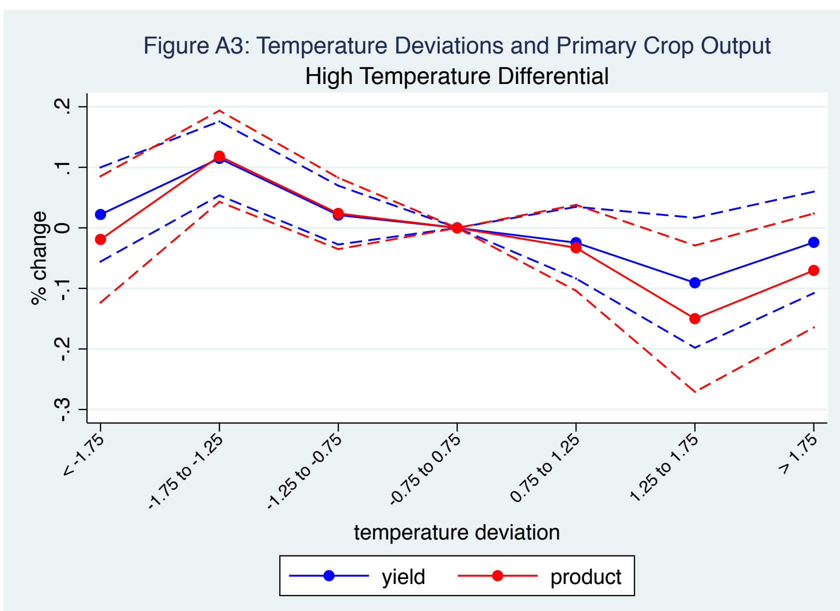

Figure A3: Interaction coefficients from a regression of the monsoon agricultural output on monsoon temperature (dds) in the indicated standard deviation bins, where the interaction terms give the differential effect for shocks in high mean temperature (dds) districts.

Table 1: Crime and Weather Shocks Across Decades

	burlgary (1)	banditry (2)	thefts (3)	robbery (4)	riots (5)	kidnap (6)	rape (7)	murder (8)
negative rainfall								
1970s	0.029	0.076*	-0.039	0.083*	0.099***	0.029	0.047	0.021
	(0.038)	(0.045)	(0.035)	(0.045)	(0.033)	(0.033)	(0.036)	(0.024)
1980s	0.068***	0.022	0.042*	0.011	0.022	-0.013	0.010	0.027*
	(0.021)	(0.035)	(0.022)	(0.031)	(0.032)	(0.025)	(0.028)	(0.016)
1990s	0.067***	0.033	0.025	0.066**	0.059**	0.059**	0.030	0.022
	(0.023)	(0.038)	(0.019)	(0.033)	(0.029)	(0.025)	(0.030)	(0.019)
positive temperature								
1970s	0.053	0.256***	0.131***	-0.017	0.129***	0.018	0.107***	0.04
	(0.039)	(0.049)	(0.035)	(0.049)	(0.041)	(0.039)	(0.040)	(0.026)
1980s	0.002	0.013	0.033	0.136**	0.114***	-0.040	0.074*	-0.002
	(0.032)	(0.046)	(0.030)	(0.054)	(0.043)	(0.040)	(0.039)	(0.021)
1990s	0.006	0.042	0.03	0.068**	0.043	0.043*	-0.007	0.036*
	(0.028)	(0.035)	(0.023)	(0.029)	(0.029)	(0.025)	(0.026)	(0.020)

The outcome variable is crimes per 100k. Regressions are estimated separately for each type of crime. The weather shocks are defined as rainfall/temperature 1 standard deviation from the mean; dummies are included separately for shocks occurring in each decade. Dummies are included for the district-crime. Year fixed effects are included, as well as state-year time trends. Error terms are clustered at the district level.